第七章 磁力

§1 带电粒子在磁场中运动

§2 电场和磁场的相对性和统一性

§3 霍尔效应

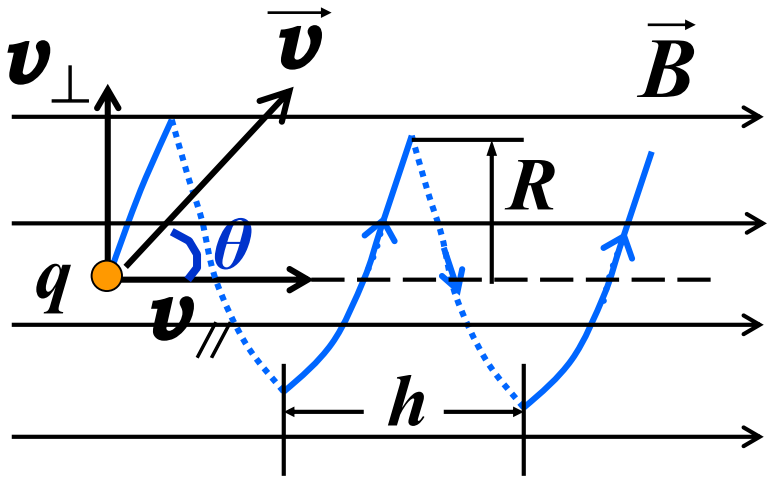
§4 安培力

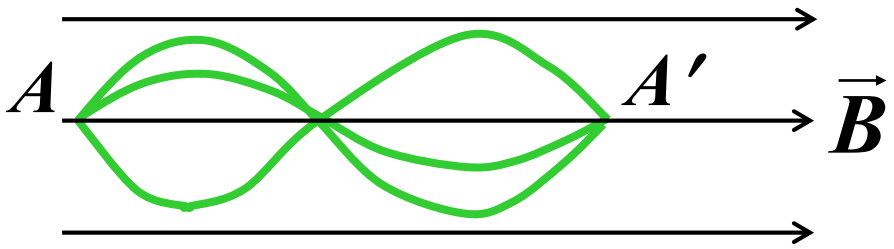
§5 载流线圈在磁场

§1 带电粒子在磁场中运动

洛仑兹力 $\boxed{\vec{f}_m = q\vec{v} \times \vec{B}}$

磁聚焦



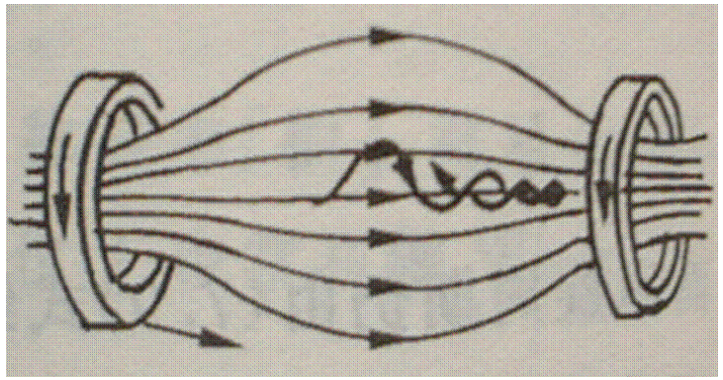


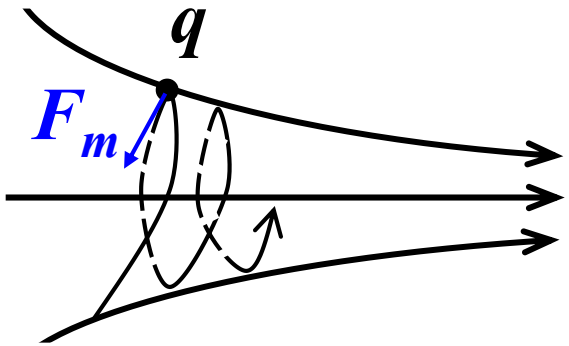
在θ不大、v几乎相同的条件下，形成磁聚焦

$$R = \frac{mv_\perp}{qB},\quad T = \frac{2\pi m}{qB},\quad h = v_{//}T = \frac{2\pi m}{qB} v\cos\theta$$

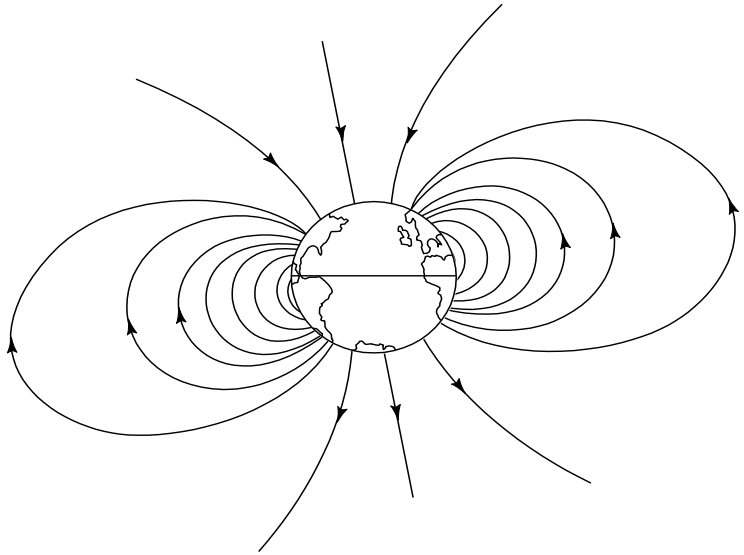
▲磁约束

磁瓶



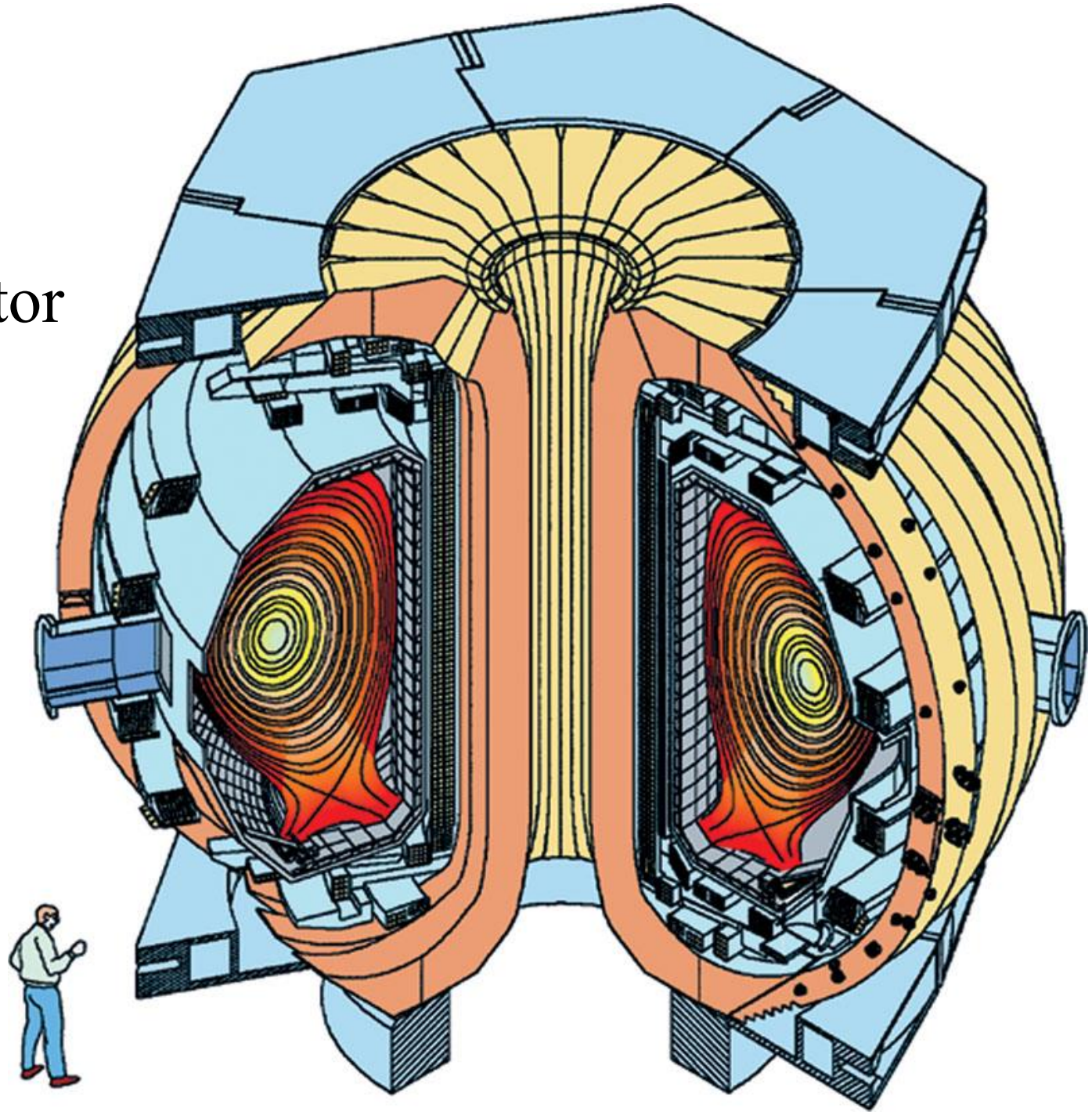


速率不变



范阿仑带

Tokamak
Fusion reactor



*绚丽多彩的极光

在地磁两极附近，由于磁感线与地面垂直，外层空间入射的带电粒子可直接射入高空大气层内，它们和空气分子碰撞产生的辐射，就形成了极光。

§2 电场和磁场的相对性和统一性

$$\boxed{\vec{f} = q\vec{E} + q\vec{\upsilon} \times \vec{B}}$$

洛仑兹力公式

某个瞬间参考系速度与电荷运动速度相同

磁力消失？在不同参考系物理结果应相同

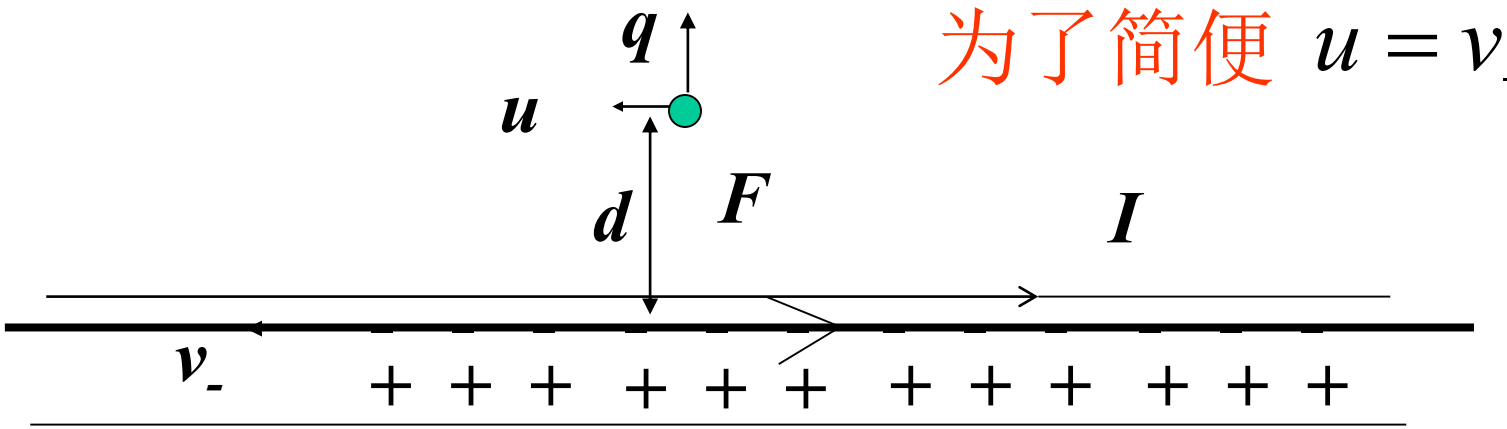
相对论角度

磁场在另一个惯性系变成电场

电场在另一个惯性系变成磁场

无限长载流导线电流 I 与一运动电荷 q 相距 d

为了简便 $u = v_-$



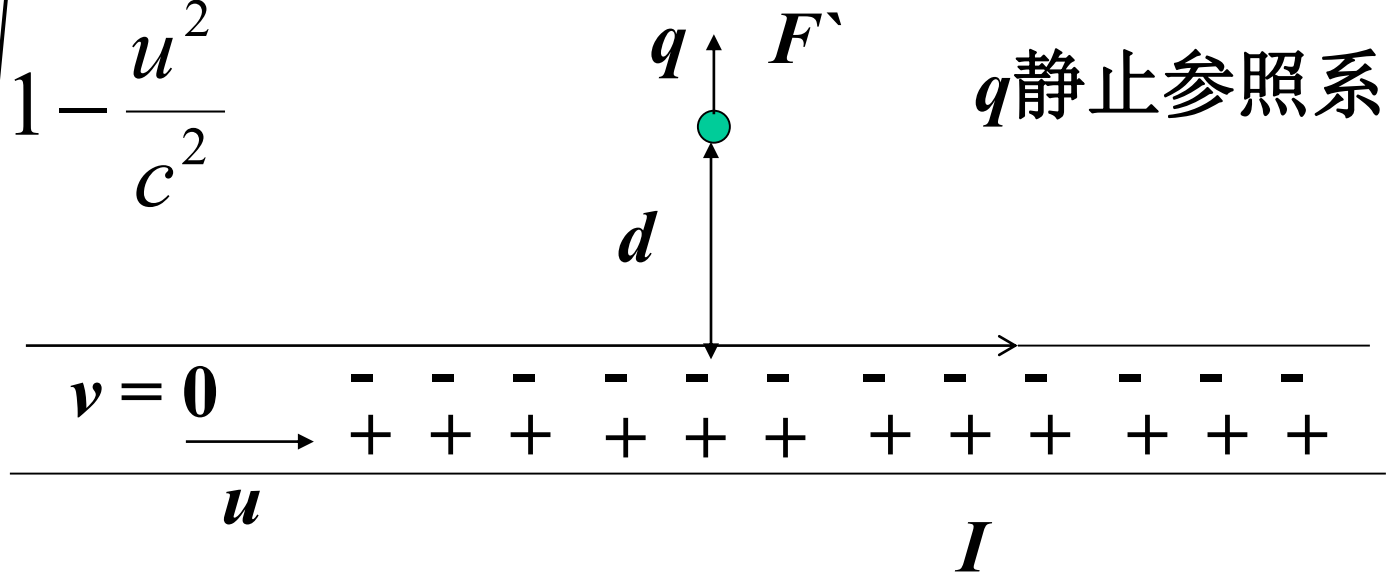
I 在 q 处 B 为

$$B = \frac{\mu_0 I}{2\pi d}$$

根据洛伦兹力，q受力

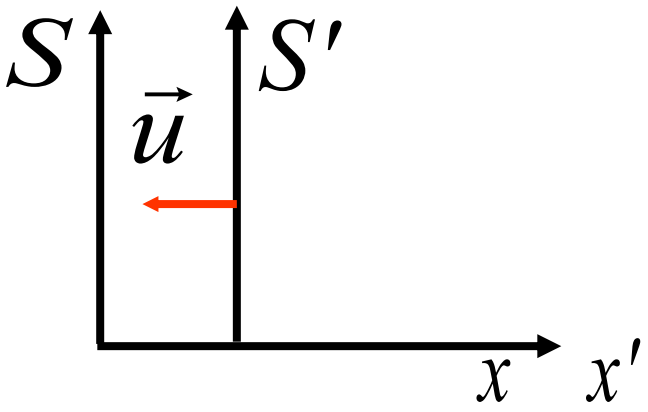
$$F_m = \frac{\mu_0 I q u}{2\pi d}$$

$$\gamma = \frac{1}{\sqrt{1-\frac{u^2}{c^2}}}$$



$$\because l = \frac{l_0}{\gamma} \qquad l_+ = \frac{l_{+0}}{\gamma}$$

$$l_- = \gamma l_{-0}$$



电荷密度反比于间距

$$\lambda_- = \frac{\lambda_0}{\gamma} \qquad \lambda_+ = \lambda_0 \gamma$$

$$F' = q\frac{\lambda_+ - \lambda_-}{2\pi\varepsilon_0 d} = \frac{\lambda_0}{2\pi\varepsilon_0 d}\gamma\frac{u^2}{c^2}q = \frac{\mu_0 I q u}{2\pi d}\gamma$$

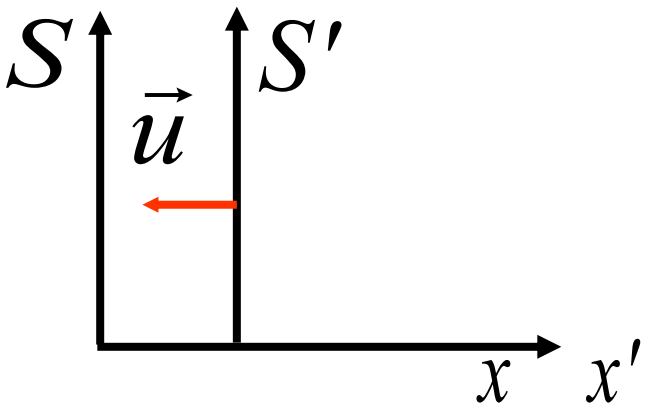
$$\boxed{F' = \gamma F}$$

$$\frac{F_y'}{F_y} = \frac{\Delta p_y'/\Delta t'}{\Delta p_y/\Delta t}$$

$\searrow \gamma$

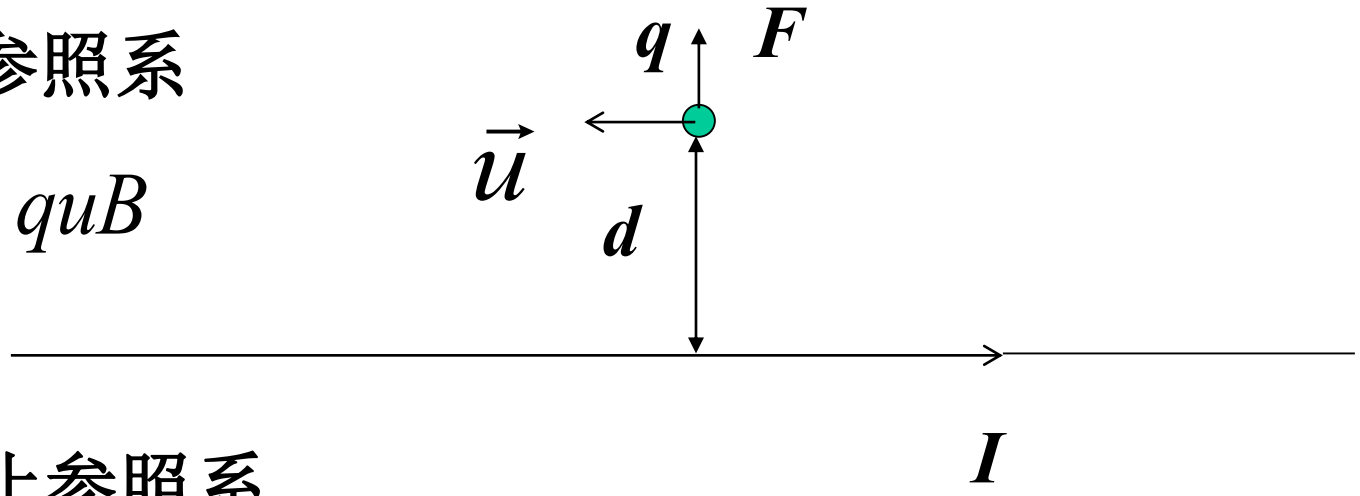
原时最短

$$\boxed{\Delta t = \gamma \Delta t'}$$



地面参照系

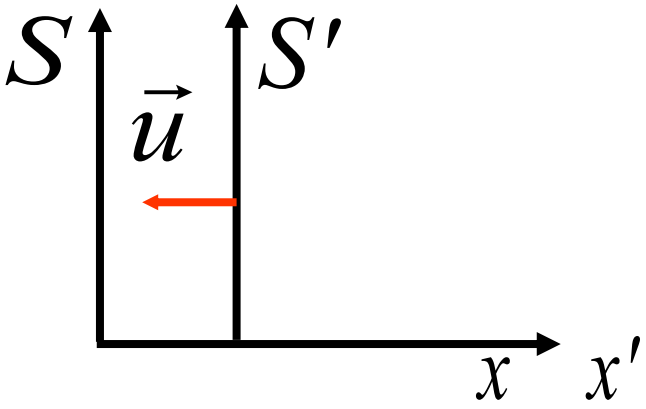
$$F = quB$$



q静止参照系

$$\gamma F = \gamma quB$$

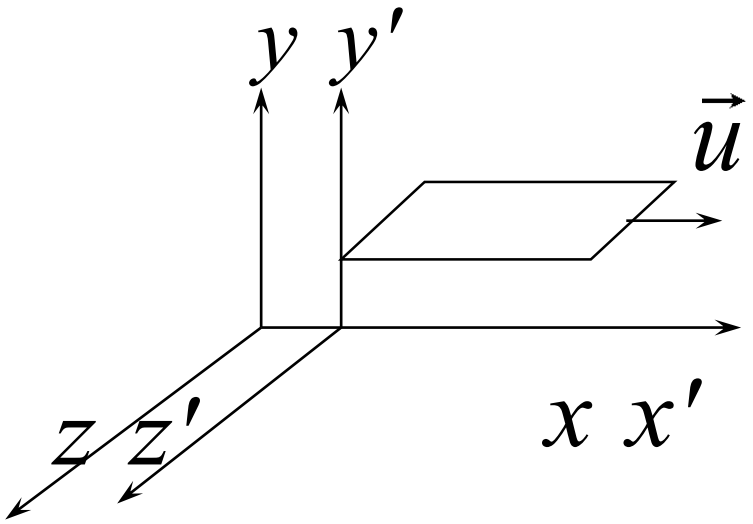
$$E'_y = \gamma uB$$



地面参考系只有磁场

运动参考系还有电场

均匀带电的无限大平面电荷面密度 σ



S` 系中只有静电场

$$\vec{E}' = \vec{E}'_y = \pm\frac{\sigma}{2\varepsilon_0}\hat{y}$$

$$S: \qquad \vec{j} = \sigma'\vec{u} = \gamma\sigma\vec{u}$$

$$\vec{B} = \pm\frac{\mu_0}{2}\gamma\sigma u\hat{z}$$

$$E_y = \gamma E'_y$$

$$\boxed{\vec{B} = \frac{\vec{u}\times\vec{E}}{c^2}}$$

$$= \pm\gamma\frac{E'_y u}{c^2}\hat{z}$$

电场—磁场似乎是完全不同

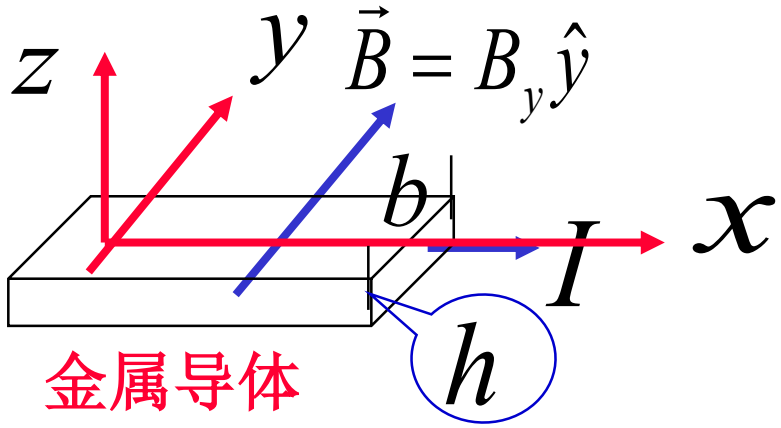
相对论的角度　电场—磁场相互紧密关联

惯性系变换可以使 电场—磁场 互相变换

电磁场方程是在惯性系成立

场线只是形象描述, 不能真把它当“线”
以为可以识别是否相对场运动

§3　霍耳效应



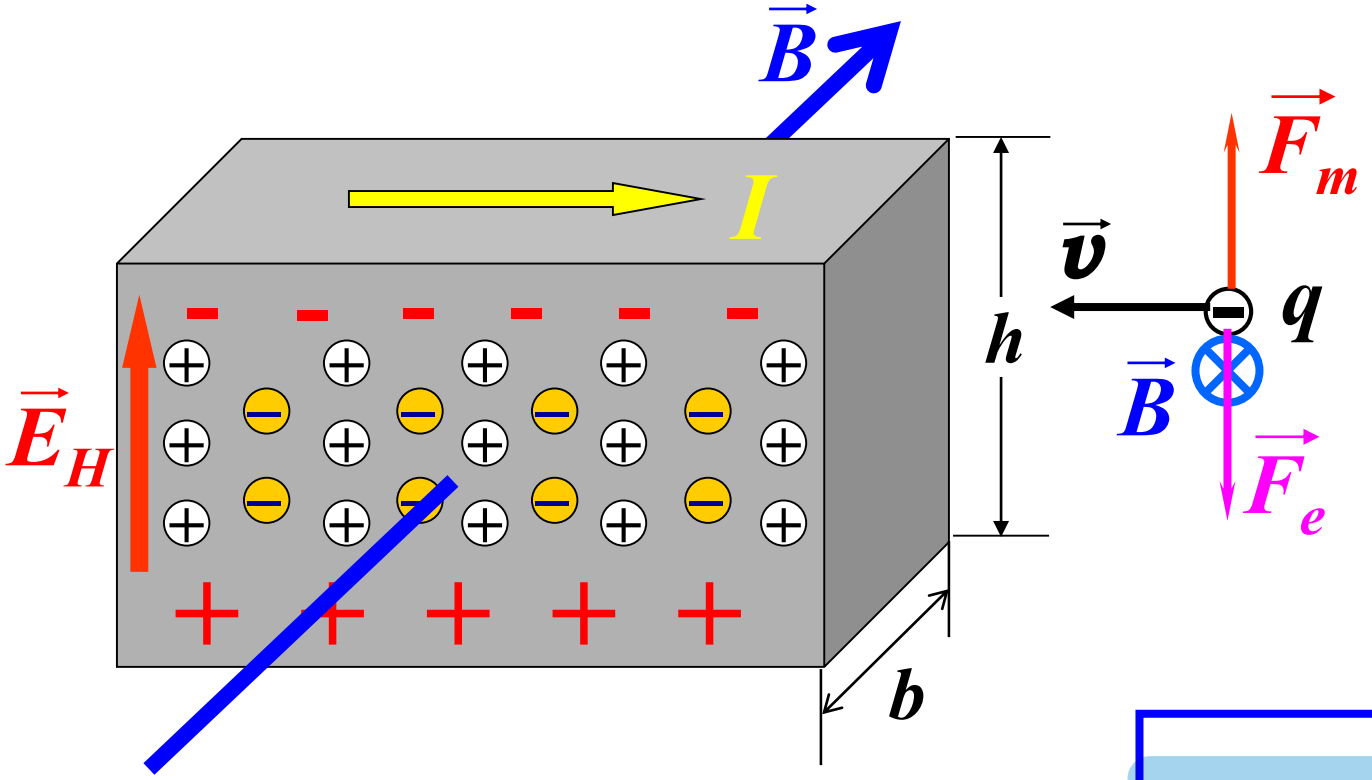
1879年美国物理学家霍耳发现：

对应图中沿 z方向有电势差

霍耳电势差：

$$U_H \propto \frac{IB}{b}$$

霍耳电势差：



$$q\upsilon B = qE_H$$

$$nq\upsilon bh = I$$

$$E_H h = U_H$$

$$\boxed{K_H = \frac{1}{nq}}$$

$$U_H = \frac{IB}{nbq} = K_H \frac{IB}{b}$$

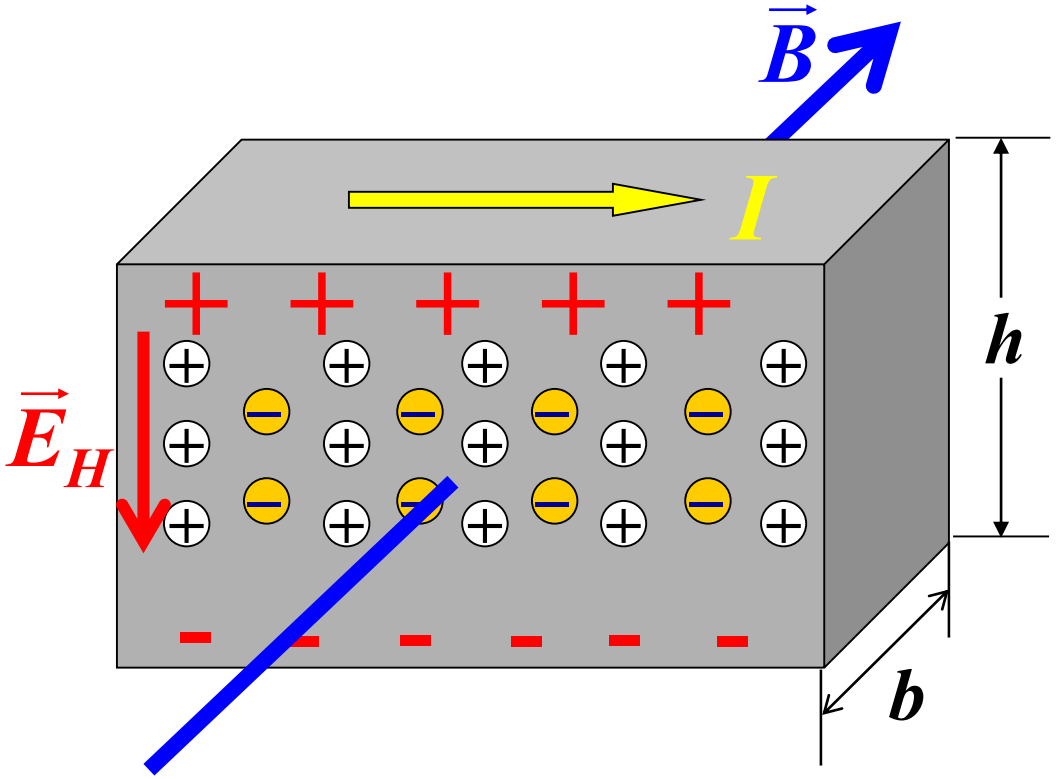
K_H —— 霍耳系数

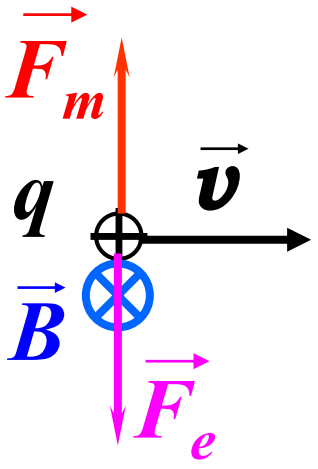
只与材料有关.

主要取决于电导率

导电性↑ n↑ K_H↓

正负⇒载流子类型 大小⇒浓度





$$U_H = \frac{IB}{nbq} = K_H \frac{IB}{b}$$

判断载流子种类

测量载流子浓度　　测磁场等

霍耳电阻

$$R_H = \frac{U_H}{I} = \frac{B}{nqb} \propto B$$

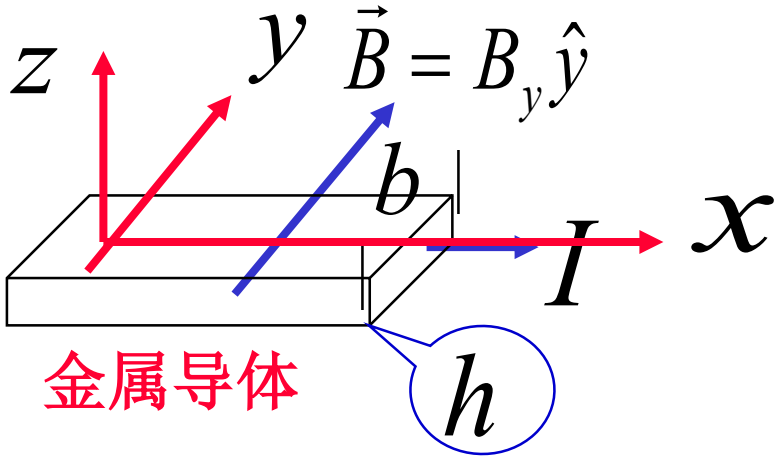
（有电阻的量纲）

此处：U与I方向垂直

霍耳效应

精确的解释只能用电子的量子理论

（包括反常情形）



*量子霍耳效应：

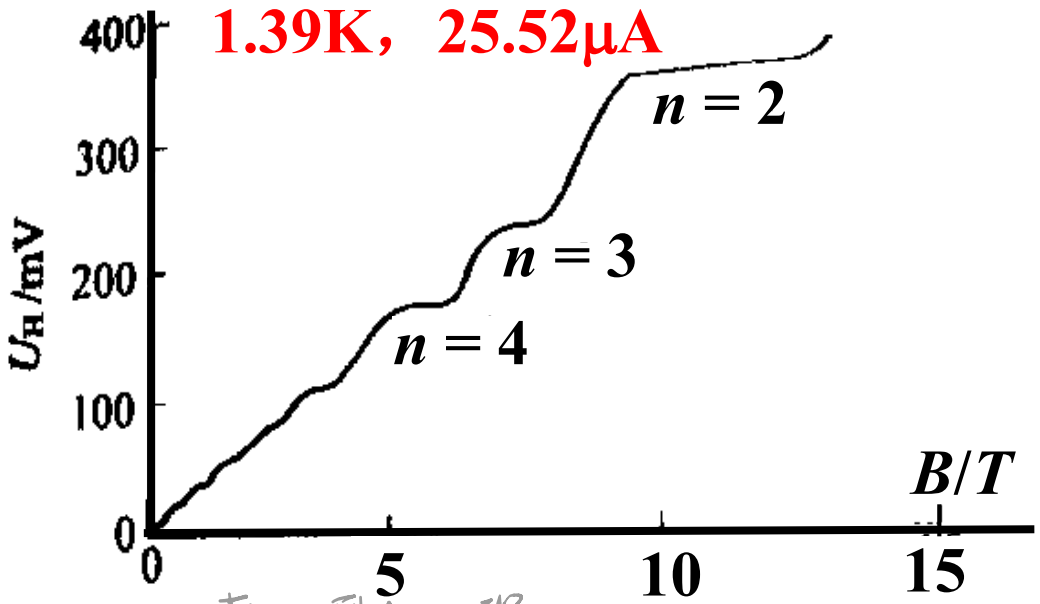
1980年克里青发现，

在极低温、强磁场下

$$R_H \not\propto B$$

$$R_H = \frac{R_K}{n}, \quad n = 1,2,3,\cdots$$

↘电荷量子化，只能以整数倍出现



克里青（Klitzing）常量 $R_K = \frac{h}{e^2} = 25812.80\Omega$

R_K的测量准确到10^{-10}

1990年定义 $1\Omega = \frac{R_K}{25812.80}$

*分数量子霍耳效应：

崔琦和施特默（Störmer）发现在更强的磁场下，n 可以是分数，如：1/3、1/5、1/2、1/4等，这称为分数量子霍耳效应。

劳克林（Laughlin）：携带分数电荷的准粒子

克里青　1985年诺贝尔物理学奖

劳克林、施特默和崔琦

1998年诺贝尔物理学奖

反常霍耳效应：

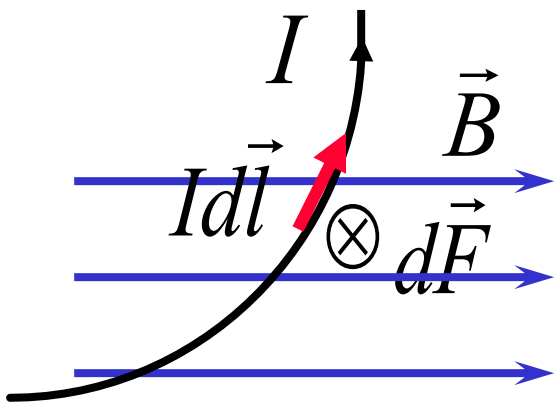
1880年霍耳发现, 不需要外加磁场, 某些材料靠自发磁化产生霍耳效应.

量子反常霍耳效应：

2013年3月15日，《科学》（Science）杂志在线发文, 清华大学物理系薛其坤和王亚愚等与科学院物理所及斯坦福大学物理学家合作, 观察到量子反常霍耳效应.

诺奖级别的研究成果

§4 安培力



怎么计算电流受到的磁场力？

电流元受力
载流子受力之和

安培指出 任意电流元受力为

$nq\langle\vec{v}\rangle\cdot dV$
$=\vec{j}\cdot dV$
$=\vec{j}\cdot S\cdot dl$
$=I\,dl$

$$\boxed{d\vec{F}=Id\vec{l}\times\vec{B}}$$

安培力公式

整个电流受力

$$\boxed{\vec{F}=\int_{(l)}Id\vec{l}\times\vec{B}}$$

例1 平行电流导线

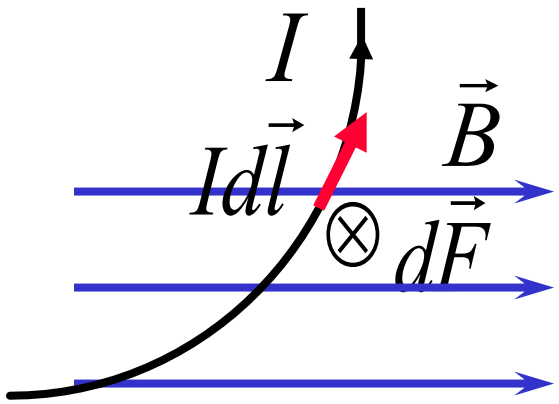
- 安培力的本质是洛仑兹力

$$d\vec{F} = Id\vec{l} \times \vec{B}$$

$$= nq\vec{\upsilon}\Delta S dl \times \vec{B}$$

$$= \vec{j} \times \vec{B} dV$$

$$= dNq\vec{\upsilon} \times \vec{B}$$

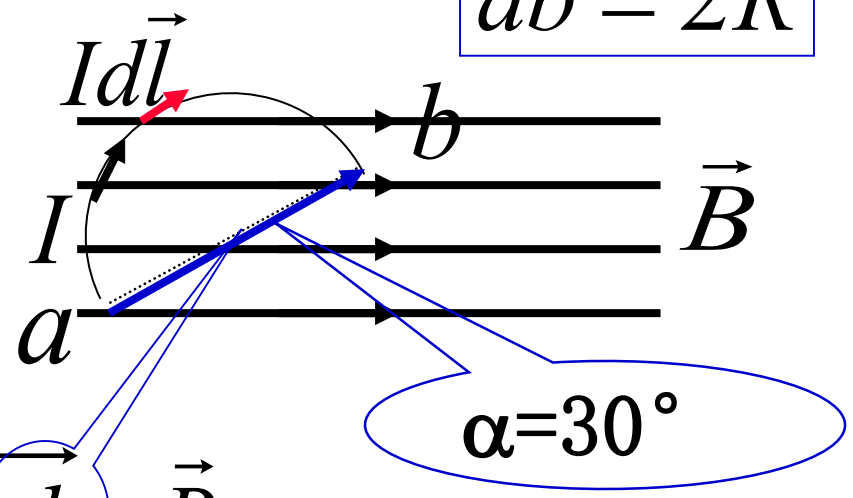


单个运动带电粒子 $q\vec{\upsilon} \to Id\vec{l}$

例2 在均匀磁场中放置一半径为R的半圆形导线，电流强度为I，导线两端连线与磁感强度方向夹角α=30°，求此段圆弧电流受的磁力。

$$\boxed{\overline{ab}=2R}$$

解：在电流上任取电流元$Id\vec{l}$



$$\vec{F}=\int_{(a)}^{(b)} Id\vec{l}\times\vec{B}$$

场均匀

$$=I\left[\int_{(a)}^{(b)} d\vec{l}\right]\times\vec{B}=I\overrightarrow{ab}\times\vec{B}$$

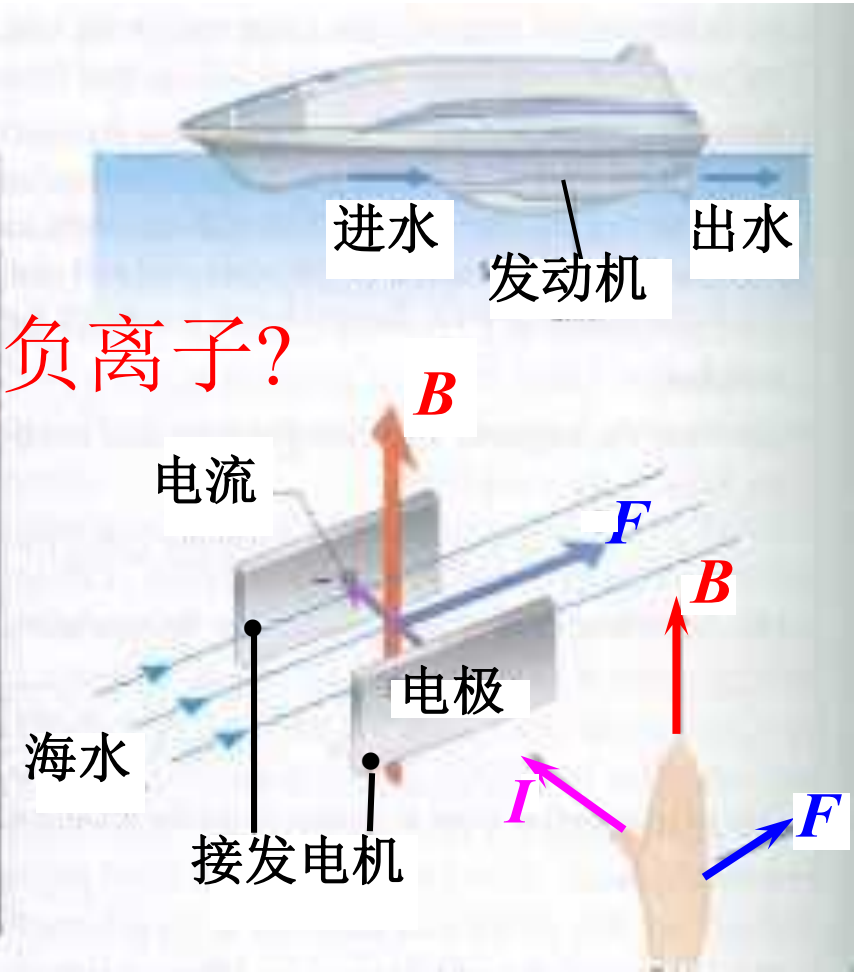
均匀场中：$\vec{B}$可以移至积分号外.

$\int_{(a)}^{(b)} d\vec{l}$ → 首尾相连的矢量

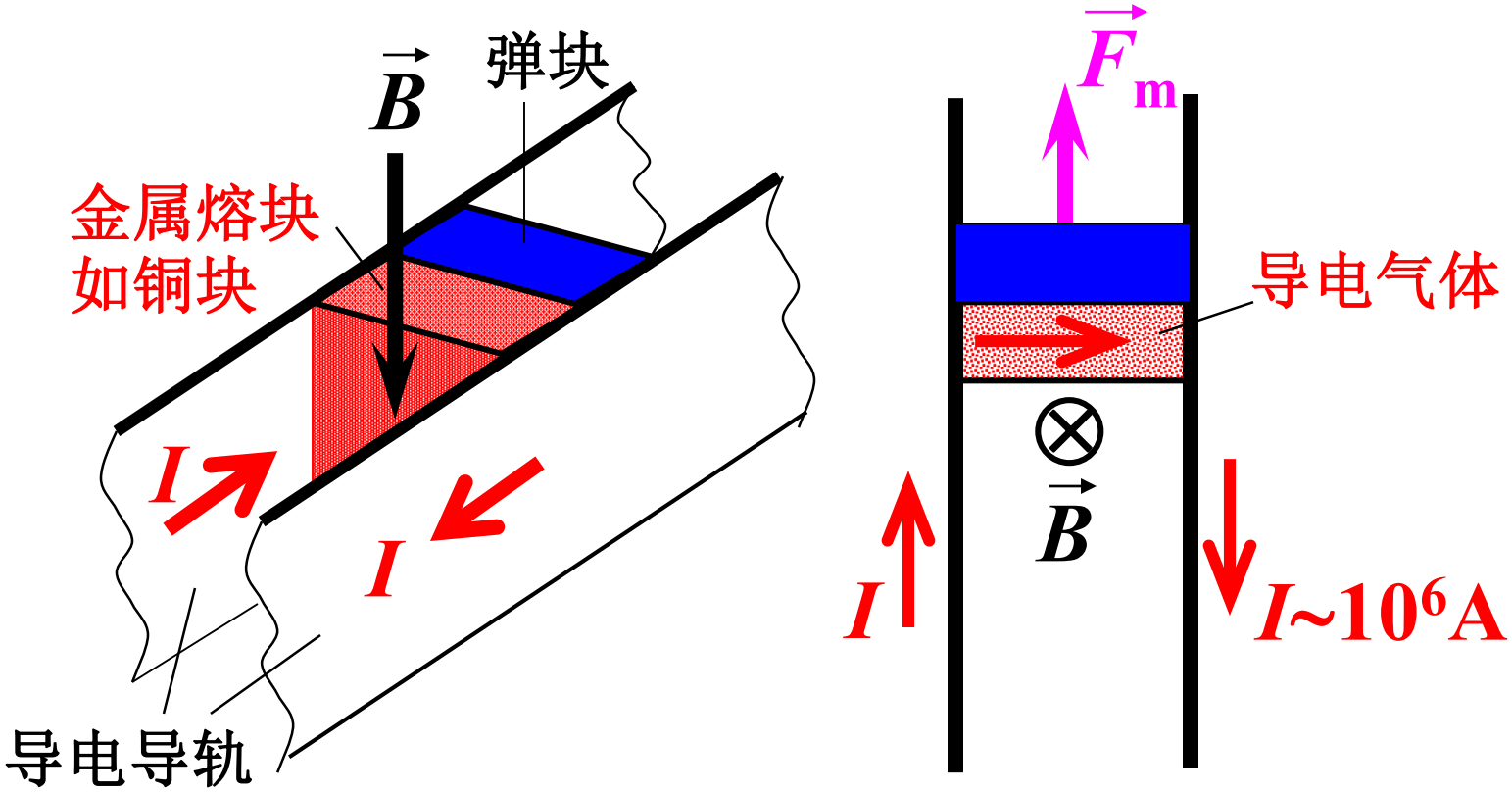
$$\left|\vec{F}\right|=I\left|\overrightarrow{ab}\right|B\sin\alpha=IBR$$

方向 $\otimes\ \vec{F}$

磁流体船



电磁轨道炮



I. B大
可施加
很大的加速度

弹块 $\bar{a} \sim 10^6 g$，在1ms内，速度可达10km/s

§5 载流线圈在磁场

载流线圈在均匀磁场中

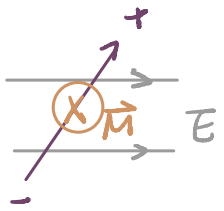
$$\vec{F} = \int_{(l)} I d\vec{l} \times \vec{B}$$

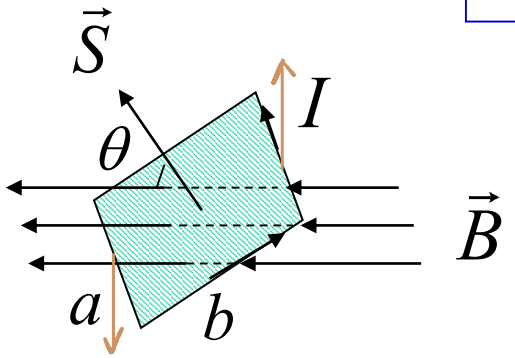
★合力 $\boxed{\vec{F}_{合} = 0}$

✸力矩 $\boxed{\vec{M} = \vec{p}_m \times \vec{B}}$ 磁矩

力矩 $\vec{M} = \vec{p} \times \vec{E}$

$W_e = -\vec{p} \cdot \vec{E}$



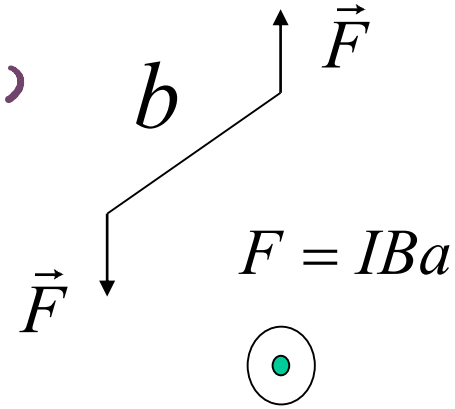


$\vec{p}_m$ 与 $\vec{S}$ 同向.(法向)

平面 // $\vec{B}$ ⇒ $\vec{p}_m \perp \vec{B}$

受 $\vec{M}$ 最大.

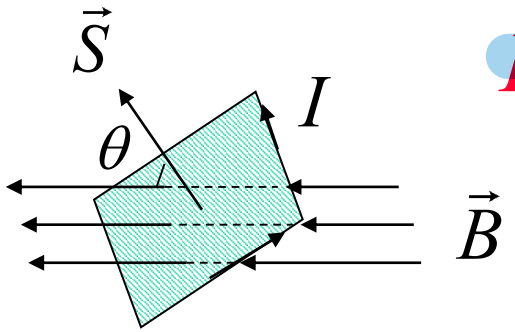
平面 $\perp \vec{B}$ ⇒ $\vec{M} = 0$



$F = IBa$

$$M = Fb\sin\theta = IBab\sin\theta = p_m B\sin\theta$$

载流线圈在均匀磁场中的能量



I 不变

$$W_m(\theta)-W_m(0)=\int_0^{\theta} M d\theta$$

$$=\int_0^{\theta} ISB\sin\theta d\theta$$

$$=-ISB(\cos\theta-\cos 0^o)$$

$$\boxed{W_m=-\vec{p}_m\cdot\vec{B}}$$

刚性:
形状、电流强度不变
(接电源)

I 可变

$$W_m=\vec{p}_m\cdot\vec{B}$$

与静电场对比

$$\vec{M}=\vec{p}_m\times\vec{B}$$

$$W_m=\pm\vec{p}_m\cdot\vec{B}$$

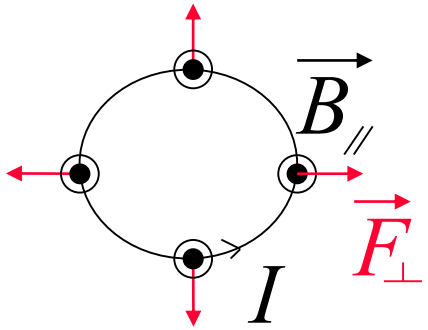
$$\vec{M}=\vec{p}_e\times\vec{E}$$

$$W_e=-\vec{p}_e\cdot\vec{E}$$

*载流线圈在非均匀磁场中受力

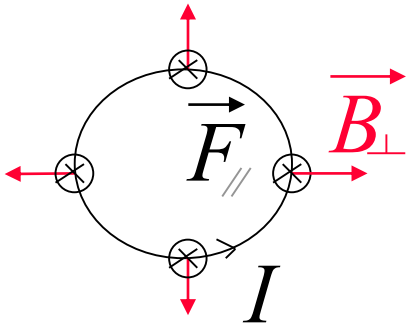
I 不变

$$\vec{B} = \vec{B}_{//} + \vec{B}_{\perp} \qquad \vec{p}_m = I\Delta S\,\hat{z}$$



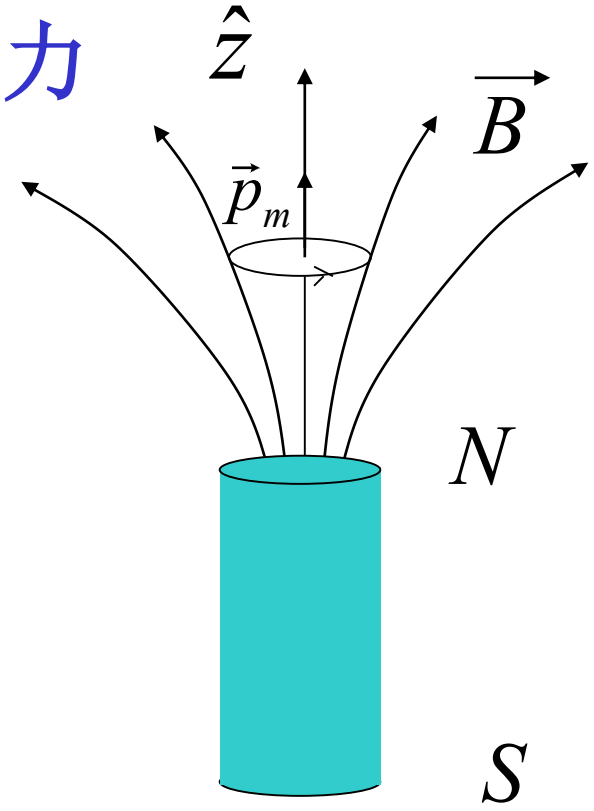
$\vec{F}_{\perp}$抵消

$$\vec{F}_{\perp} = 0$$



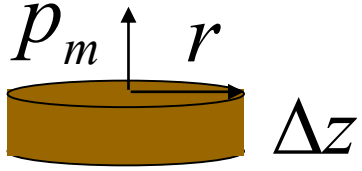
合力向下吸引

$$\vec{F}_{//} = -2\pi r I B_{\perp}\hat{z}$$



受吸引力.

看作一个小磁针



B场在z方向减少，x方向增加

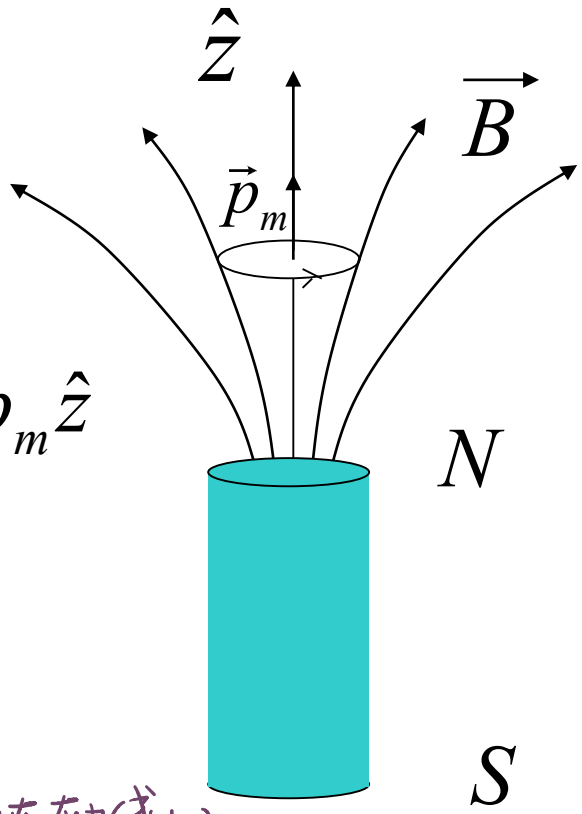
$$\oiint_S \vec{B}\cdot d\vec{S} = 0 \quad \Rightarrow \quad [B_{//}(z+\Delta z) - B_{//}(z)]\Delta S + B_{\perp}2\pi r\Delta z = 0$$

$$[B_{//}(z+\Delta z)-B_{//}(z)]\Delta S+B_{\perp}2\pi r\Delta z=0$$

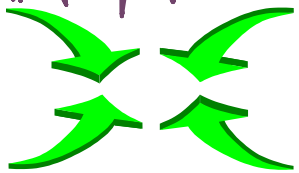
$$2\pi rB_{\perp}=-\frac{\partial B_{//}}{\partial z}\Delta S$$

$$\vec{F}_{//}=-2\pi rIB_{\perp}\hat{z}=\frac{\partial B_{//}}{\partial z}I\Delta S\hat{z}=\frac{\partial B_{//}}{\partial z}p_m\hat{z}$$

$$F_z\hat{z}=p_m\frac{\partial B_{//}}{\partial z}\hat{z}=\vec{p}_m\cdot\vec{\nabla}B_z\hat{z}$$



(放在轴线上)

$$\boxed{\vec{F}=\vec{p}_m\cdot\vec{\nabla}\vec{B}}$$

能判断吸引/排斥即可